

Teaching Statement

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➤ Teaching Philosophy

My goal as an educator is to guide students to become independent thinkers, engineers and researchers who can **think, implement, and reason about robotic and autonomous systems in near-future world**. I emphasize project-driven, personalized learning that tightly integrates theory, computation, and physical implementation. My teaching philosophy is built on three core principles.

1. Project-driven and design-oriented learning

I organize courses around open-ended engineering problems rather than isolated theoretical exercises. Students are guided to formulate problems, design system architectures, implement algorithms, and evaluate performance through simulation and hardware experiments. For example, in an Autonomous Flight course, a capstone project may involve designing a UAV system capable of autonomous takeoff, trajectory tracking, obstacle avoidance, safe landing, integrating dynamics, control, perception, and software systems.

2. Strong integration of theory and practice

I emphasize that rigorous theory underpins robust and safe autonomous systems. My courses therefore integrate mathematical modeling, control theory, and optimization with hands-on implementation using MATLAB/Simulink, ROS-based simulators, and robotic platforms. Students see how analytical models translate directly into code, simulation results, and real experimental data, reinforcing conceptual understanding and engineering intuition.

3. Interdisciplinary and systems thinking

Robotics and autonomous systems inherently combine mechanical design, electronics, control, AI, and human factors. I encourage students to think holistically about system constraints, safety, reliability, and ethical considerations. This approach is particularly important for applications such as UAVs, field robotics, and multi-robot systems, which operate in uncertain, safety-critical environments.

➤ Teaching Experience

At Nanyang Technological University (NTU), I have contributed to both specialized robotics courses and interdisciplinary AI education. I have served as Teaching Assistant or instructor for courses including **Autonomous Flight**, **Introduction to Aerodynamics**, and **Air Traffic Management and Operations**. My responsibilities included delivering tutorials, designing laboratory exercises, supervising projects, and mentoring students. These experiences

strengthened my ability to teach mathematically intensive material while maintaining strong connections to practical applications in aerial and autonomous systems.

In addition, I have been involved in the **university-wide course Artificial Intelligence Literacy**, which introduces graduate students from diverse disciplines to AI fundamentals, applications, ethics, and societal impact. I facilitated project-based learning, guided interdisciplinary teams, and assessed both technical reasoning and communication. This experience prepares me well to teach AI-integrated literacy content to students with varied academic backgrounds. I received several students' excellent feedback. This experience is directly relevant to **student-centered, active learning** and its desire for faculty who can teach AI-related content in a way that is accessible across disciplines while still rigorous.

➤ **Mentorship and Student Development**

I view mentorship as an essential component of teaching. I have supervised and co-supervised 4 PhD, 4 Master's, and 20+ undergraduate students on topics including multi-robot systems, drone swarms, reinforcement learning, and vision-based autonomy. I emphasize research rigor, experimental design, clear communication, and independent problem-solving. Many of the students I mentored progressed to research publications, competitions, or advanced graduate study.

➤ **Vision for Future Teaching**

I envision my teaching as an ongoing journey of innovation and adaptation. I plan to contribute through the following teaching areas.

- **Core and advanced robotics courses**, such as Autonomous Flight, Aerial Robotics, Robotics and Mechatronics, and Multi-Robot Systems, with strong emphasis on project-based learning and system integration.
- **AI-enabled robotics courses**, covering learning-based control, perception, and decision-making, as well as the integration of robotics and AI for UAV, UGV, and field robotics applications.
- **Interdisciplinary project modules** that connect robotics with sensing, AI, human-robot interaction, and real-world applications in urban, offshore, agricultural, and infrastructure settings.

I also plan to incorporate **active learning methods** such as guided coding sessions, design reviews, peer feedback, and reflection on safety and ethical considerations in robotics. I am committed to fostering a culture of inclusion and equity, inspiring students to think critically, work collaboratively, contribute meaningfully to society and feel empowered to succeed, regardless of their backgrounds.